
REPRESENTATION OF DISCRETE MAXIMUM ENTROPY DISTRIBUTIONS AS TENSOR NETWORKS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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ABSTRACT

This work investigates discrete distributions, which maximize the entropy with respect to linear constraints by the expectation of statistics. We introduce Computation-Activation Networks, which are tensor network formats representing generic maximum entropy distribution. The characterization exploits the face lattice of corresponding mean parameter polytopes to statistics, depending on which we provide rank bounds of the tensor network format. For boolean statistics we provide an interpretation based on propositional logics, which we utilize to construct statistics of generic lattice. We furthermore show that the empirical means are minimal sufficient statistics for the estimation of maximum entropy distributions.

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1 Introduction

The Shannon entropy of a distribution quantifies the amount of structure, or information, and appears therefore as an important concept across many scientific areas. By Shannons source code theorem Shannon (1948), entropy describes fundamental limits of information transfer. In statistical physics, founded on the ergodic assumptions, maximum entropy distributions such as Maxwell-Boltzmann distributions are the gold standard to model equilibrium distributions.

The principle of entropy maximization also appears naturally in learning problems. Extracting characteristics observed in data does not determine an estimated model yet. Choosing the distribution with the least structure among those reflecting the observed characteristics is a regularization principle. Here one can choose the Shannon entropy to quantify the negative amount of structure in the distribution and maximize the quantity with respect to the constraints. Furthermore, maximum entropy problems are the dual problems to convex maximum likelihood problems.

Less obvious, studying generalizing maximum entropy distributions might be a key step towards a mathematically sound unification of logical and probabilistic models. Combining logical and probabilistic AI is a necessity in statistical relational AI, neuro-symbolic AI and explainable AI. Probability theory with exponential families typically focuses on distributions with maximal support. Logics on the contrary can be understood as a study of the support of models, marking the worlds consistent with a knowledge base. Logical and probabilistic models can be treated in the tensor network formalism, which thus serves as a unifying representation language.

Maximum entropy principles appear further in reinforcement learning Yoon et al. (2024); Song et al. (2025), AGI Miotto and Monacelli (2025) and Bayesian NN Rathnakumar et al. (2026).

1.1 Tensor Notation

We introduce tensors based on index variables X_k to each of the $k \in [d] = \{0, \dots, d-1\}$ axes, each taking values in $[m_k] := \{0, \dots, d-1\}$, where $m_k \in \mathbb{N}$ is the dimension of the variable. Tensors are then maps

$$\tau [X_{[d]}] : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}$$

from indices $x_{[d]} = (x_0, \dots, x_{d-1}) \in \bigtimes_{k \in [d]} [m_k]$ to their coordinates $\tau [X_{[d]} = x_{[d]}] \in \mathbb{R}$. One-hot encodings of index tuples $x_{[d]}$ are specific tensors $\epsilon_{x_{[d]}} [X_{[d]}]$ such that $\epsilon_{x_{[d]}} [X_{[d]} = \tilde{x}_{[d]}] = 1$ if $\tilde{x}_{[d]} = x_{[d]}$ and 0 else.

Given a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ of an arbitrary node set $\mathcal{V}$ and hyperedges $e \in \mathcal{E}$ being subsets of $\mathcal{V}$, we define decorating tensor network as follows. Each node $v \in \mathcal{V}$ gets decorated by a variable X_v of dimension m_v and each hyperedge $e \in \mathcal{E}$ by a tensor $\tau^e [X_e]$ with the index variables to the contained nodes. The set of decorating tensors then builds the tensor network $\tau^{\mathcal{G}}$ on the hypergraph. A contraction of the tensor network keeping the nodes $\mathcal{U} \subset \mathcal{V}$

open is then the tensor $\langle \tau^G \rangle_{[X_{\mathcal{U}}]}$ with variables $X_{\mathcal{U}}$ and coordinates to the indices $x_{\mathcal{U}}$ by

$$\langle \tau^G \rangle_{[X_{\mathcal{U}}=x_{\mathcal{U}}]} = \sum_{x_{\mathcal{V} \setminus \mathcal{U}}} \prod_{e \in \mathcal{E}} \tau^e [X_e = x_e] .$$

1.2 The Maximum Entropy Problem

Given a non-negative and non-vanishing base measure $\nu [X_{[d]}]$, a probability distribution is a non-negative tensor $\mathbb{P} [X_{[d]}]$ such that

$$\langle \mathbb{P} [X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]} = 1 .$$

We denote the set of such distributions by

$$\begin{aligned} \Lambda^\nu = \{ & \mathbb{P} [X_{[d]}] : \mathbb{P} [X_{[d]}] \geq 0 [X_{[d]}], \langle \mathbb{P} [X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]} = 1, \\ & \forall x_{[d]} \in \times_{k \in [d]} [m_k] \nu [X_{[d]} = x_{[d]}] = 0 \Rightarrow \mathbb{P} [X_{[d]} = x_{[d]}] = 0 \} . \end{aligned}$$

Note that we here restrict the support of the tensor $\mathbb{P} [X_{[d]}]$ to be included in the support of the base measure, to avoid ambiguity in the representation.

The entropy of a distribution is

$$\mathbb{H}_\nu [\mathbb{P} [X_{[d]}]] = - \langle \mathbb{P} [X_{[d]}], \ln [\mathbb{P} [X_{[d]}]], \nu [X_{[d]}] \rangle_{[\emptyset]} .$$

Note that the entropy depends on the chosen base measure.

The mean parameter of a distribution $\mathbb{P} [X_{[d]}]$ to a statistic $t : \times_{k \in [d]} [m_k] \rightarrow \times_{s \in [n]} [p_s]$ is the vector $\mu [L] \in \mathbb{R}^p$ with the coordinates

$$\mu [L = \ell] = \mathbb{E} [f_\ell] = \langle f_\ell [X_{[d]}], \mathbb{P} [X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]} .$$

We express the computation of the mean parameter in the contraction of the selection encoding $\sigma^t [X_{[d]}, L]$ of t

$$\mu [L] = \langle \mathbb{P} [X_{[d]}], \sigma^t [X_{[d]}, L] \rangle_{[L]} .$$

The maximum entropy problem given a mean parameter $\mu^* [L]$ is stated by

$$\operatorname{argmax}_{\mathbb{P} [X_{[d]}] \in \Lambda^\nu} \mathbb{H}_\nu [\mathbb{P} [X_{[d]}]] \quad \text{subject to} \quad \langle \mathbb{P} [X_{[d]}], \sigma^t [X_{[d]}, L], \nu [X_{[d]}] \rangle_{[L]} = \mu^* [L] \quad (\text{P}_{t, \mu, \nu})$$

1.3 Computation-Activation Networks

We now introduce with Computation-Activation Networks families of tensor networks representing, as we will show in this work, the solution of any instance of Problem $\text{P}_{t, \mu, \nu}$. Given a statistic $t : \times_{k \in [d]} [m_k] \rightarrow \times_{s \in [n]} [p_s]$ we define its basis encoding tensor

$$\beta^t [Y_{[p]}, X_{[d]}] = \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \epsilon_{t(x_{[d]})} [Y_{[p]}] \otimes \epsilon_{x_{[d]}} [X_{[d]}] .$$

A computation network is any representation of $\beta^t [Y_{[p]}, X_{[d]}]$ as a tensor network. These can be constructed in the case statistics being a composition of connective functions.

An activation tensor is $\tau [Y_{[p]}]$ and the corresponding Computation-Activation Network with respect to the statistic t and the base measure $\nu [X_{[d]}]$ the tensor

$$\mathbb{P} [X_{[d]}] = \frac{\langle \beta^t [Y_{[p]}, X_{[d]}], \tau [Y_{[p]}] \rangle_{[X_{[d]}]}}{\langle \beta^t [Y_{[p]}, X_{[d]}], \tau [Y_{[p]}], \nu [X_{[d]}] \rangle_{[\emptyset]}} .$$

We are interested in decomposition formats of $\tau [Y_{[p]}]$, where we use sets of tensor networks $\mathcal{T}^G$ on a hypergraph $\mathcal{G}$.

Definition 1. The family of by t and a $\mathcal{G}$ computable distributions are

$$\Lambda^{t, G, \nu} = \left\{ \frac{\langle \tau [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \rangle_{[X_{[d]}]}}{\langle \tau [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]}} : \tau [Y_{[p]}] \in \mathcal{T}^G \right\} .$$

1.4 Contributions

We in this paper provide tensor network representations

- Representation of distributions with maximum entropy, in case of positive realizability: Elementary activation tensors
- Representation of generic maximum entropy distributions: CP activation tensors.

Now, we want to characterize the CP rank of the activation tensors

- Depends on the face of the mean polytope, which contains the mean parameter
- We have thus a well-defined "CP rank" of faces
- Largest faces and vertices have always CP rank of 1, intermediate faces can have larger CP rank

For boolean statistics we further provide insights for boolean statistics (see Chapter 8.5):

- Example of independent statistics (see Exa. 8.28): Always elementary activation tensors (hypercubes)
- Example of partition statistics (see Exa. 8.30):
- Generic criterion for elementary activation: "Cube-like" polytopes (see Def. 8.29)

1.5 Outlook

To prepare for the presentation of our main results we introduce

- Mean polytopes: Polytopes, which contain all realizable mean parameter vectors.
- Exponential families: Maximum entropy distributions in the interior

We will then show, that dependent on the position of the mean parameter in the mean polytope, we can characterize the corresponding maximum entropy distribution by a Computation-Activation Network.

2 The mean polytope

The mean polytope is the set of possible mean parameters given the by a base measure representable distributions. We define it as

$$\mathcal{M}_{t,\nu} = \left\{ \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]} : \mathbb{P}[X_{[d]}] \in \Lambda^\nu \right\},$$

where we denote by Λ^ν the set of all probability distributions representable with respect to ν , and by $\sigma^t[X_{[d]}, L]$ the selection encoding of the statistic (see introduction).

2.1 Convex hull

First of all, we characterize the mean polytope as the convex hull of the slices to the selection encoding.

Lemma 1. *For any statistic t and base measure ν , the set of mean parameters is the convex hull of the set*

$$\mathcal{N}_{t,\nu} = \{ \sigma^t[X_{[d]} = x_{[d]}, L] : x_{[d]} \in \bigtimes_{k \in [d]} [m_k], \nu[X_{[d]} = x_{[d]}] \neq 0 \}$$

that is $\mathcal{M}_{t,\nu} = \text{conv}(\mathcal{N}_{t,\nu})$.

Proof. This follows from

$$\Lambda^\nu = \text{conv} \left(\frac{1}{\nu[X_{[d]} = x_{[d]}]} \cdot \epsilon_{x_{[d]}}[X_{[d]}] : x_{[d]} \in \bigtimes_{k \in [d]} [m_k], \nu[X_{[d]} = x_{[d]}] \neq 0 \right).$$

and for any vertex of this simplex we have

$$\left\langle \frac{1}{\nu[X_{[d]} = x_{[d]}]} \cdot \epsilon_{x_{[d]}}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \right\rangle_{[L]} = \sigma^t[X_{[d]} = x_{[d]}, L].$$

□

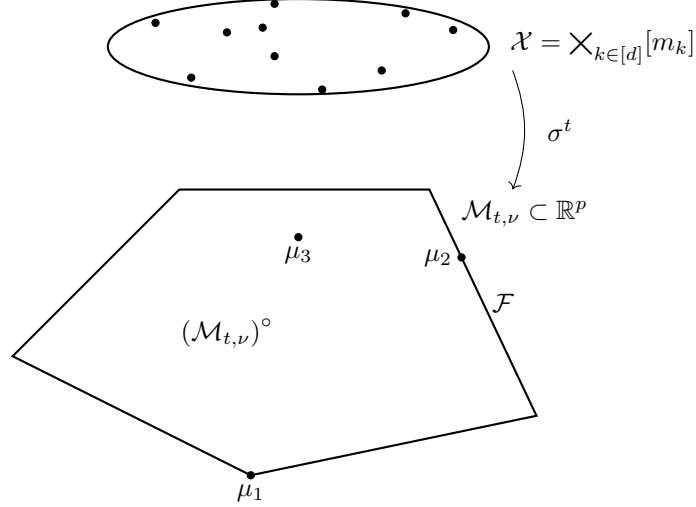


Figure 1: Sketch of the statistic mapping the states to the mean parameters.

2.2 Faces

Let us now continue with the investigation of the faces of the mean parameter polytope.

Definition 2. We say that the convex hull of a subset $\mathcal{N}_{t,\nu}^{\mathcal{F}} \subset \mathcal{N}_{t,\nu}$ is a face of a mean polytope $\mathcal{M}_{t,\nu}$, if and only if there is a face normal vector $\theta_{\mathcal{F}}[L] \in \mathbb{R}^p$ such that

$$\mathcal{N}_{t,\nu}^{\mathcal{F}} = \operatorname{argmax}_{\mu[L] \in \mathcal{N}_{t,\nu}} \langle \mu[L], \theta_{\mathcal{F}}[L] \rangle_{[\emptyset]}.$$

We denote the face as $\mathcal{F} = \operatorname{conv}(\mathcal{N}_{t,\nu}^{\mathcal{F}})$. The set of all faces of the mean polytope $\mathcal{M}_{t,\nu}$ is denoted by $L(\mathcal{M}_{t,\nu})$.

$L(\mathcal{M}_{t,\nu})$ is called a lattice Ziegler (2013).

We notice, that each face itself is a convex polytope. What is more, we can characterize these as mean parameter polytopes with respect to refined base measures to be defined next.

Definition 3. The to a face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ refined base measure is the tensor $\nu^{t,\mathcal{F}}[X_{[d]}]$ with coordinates to $x_{[d]} \in \times_{k \in [d]} [m_k]$ by

$$\nu^{t,\mathcal{F}}[X_{[d]}] = \begin{cases} \nu[X_{[d]} = x_{[d]}] & \text{if } t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}} \\ 0 & \text{else} \end{cases}.$$

We now specify the mean parameter polytope to any face using the face measure as a refinement of the base measure.

Lemma 2. For any face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ we have

$$\mathcal{F} = \mathcal{M}_{t,\nu^{t,\mathcal{F}}}.$$

Proof. For any $x_{[d]} \in \times_{k \in [d]} [m_k]$ we have $\nu^{t,\mathcal{F}}[X_{[d]} = x_{[d]}] \neq 0$ if and only if $t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}}$ and $\nu[X_{[d]} = x_{[d]}] \neq 0$. Thus we have

$$\begin{aligned} \mathcal{F} &= \operatorname{conv}(\mathcal{N}_{t,\nu}^{\mathcal{F}}) \\ &= \operatorname{conv}(\sigma^t[X_{[d]} = x_{[d]}, L] : t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}}, \nu[X_{[d]} = x_{[d]}] \neq 0) \\ &= \operatorname{conv}(\sigma^t[X_{[d]} = x_{[d]}, L] : \nu^{t,\mathcal{F}}[X_{[d]} = x_{[d]}] \neq 0) \\ &= \mathcal{M}_{t,\nu^{t,\mathcal{F}}}. \end{aligned}$$

□

Representability of a distribution with respect to face measures is an equivalent condition for the mean parameter of a distribution to be on a face, as we show next.

Lemma 3. *If and only if for a distribution $\mathbb{P} [X_{[d]}] \in \Lambda^\nu$ and a face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ we have*

$$\langle \sigma^t [X_{[d]}, L], \mathbb{P} [X_{[d]}], \nu [X_{[d]}] \rangle_{[L]} \in \mathcal{F},$$

then

$$\mathbb{P} [X_{[d]}] \in \Lambda^{\nu^{t,\mathcal{F}}}.$$

Proof. For a distribution $\mathbb{P} [X_{[d]}] \in \Lambda^\nu$ to be in $\Lambda^{\nu^{t,\mathcal{F}}}$, it is enough to show that $\mathbb{P} [X_{[d]}]$ is only supported at $\nu^{t,\mathcal{F}} [X_{[d]}]$. We have

$$\mu [L] = \sum_{x_{[d]}} \mathbb{P} [X_{[d]} = x_{[d]}] \cdot \nu [X_{[d]} = x_{[d]}] \cdot \gamma^t [X_{[d]} = x_{[d]}, L].$$

Let now $\theta_{\mathcal{F}} [L]$ be a face normal to the face $\mathcal{F}$. We then have

$$\langle \mu [L], \theta_{\mathcal{F}} [L] \rangle_{[\emptyset]} = \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \mathbb{P} [X_{[d]} = x_{[d]}] \cdot \nu [X_{[d]} = x_{[d]}] \cdot \langle \gamma^t [X_{[d]} = x_{[d]}, L], \theta_{\mathcal{F}} [L] \rangle_{[\emptyset]}.$$

Now if and only if $\mathbb{P} [X_{[d]} = x_{[d]}] \cdot \nu [X_{[d]} = x_{[d]}]$ is supported only for $x_{[d]}$ with $t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}}$ we have that

$$\langle \mu [L], \theta_{\mathcal{F}} [L] \rangle_{[\emptyset]} = \max_{\mu [L] \in \mathcal{M}_{t,\nu}} \langle \mu [L], \theta_{\mathcal{F}} [L] \rangle_{[\emptyset]}$$

which is equal to $\mu [L] \in \mathcal{F}$. Thus, if and only if $\mu [L] \in \mathcal{F}$ then $\mathbb{P} [X_{[d]}]$ is only supported at $x_{[d]}$ in the support of $\nu^{t,\mathcal{F}} [X_{[d]}]$. □

Let us now investigate tensor network representations of refined base measures, based on the basis encoding β^t of a statistic.

Theorem 1. *For any face $\mathcal{F}$ of $\mathcal{M}$ we have*

$$\nu^{t,\mathcal{F}} [X_{[d]}] = \langle \beta^t [Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}} [Y_{[p]}], \nu [X_{[d]}] \rangle_{[X_{[d]}]}$$

where

$$\kappa^{\mathcal{F}} [Y_{[p]}] = \sum_{\mu \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_{\mu} [Y_{[p]}].$$

Proof. For any $\mu \in \mathcal{N}_{t,\nu}^{\mathcal{F}}$ the tensor

$$\tau^{\mu} [X_{[d]}] = \langle \beta^t [Y_{[p]}, X_{[d]}], \epsilon_{\mu} [Y_{[p]}] \rangle_{[X_{[d]}]}$$

is the indicator of the preimage of μ under σ^t . Since preimages the elements in $\mathcal{N}_{t,\nu}^{\mathcal{F}}$ are disjoint, the support of $\tau^{\mu} [X_{[d]}]$ is disjoint and their sum

$$\sum_{\mu \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \tau^{\mu} [X_{[d]}]$$

is the indicator of the preimage of $\mathcal{F}$ under σ^t . The face measure obeys thus

$$\begin{aligned} \nu^{t,\mathcal{F}} [X_{[d]}] &= \left\langle \left(\sum_{\mu \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \tau^{\mu} [X_{[d]}] \right), \nu [X_{[d]}] \right\rangle_{[X_{[d]}]} \\ &= \sum_{\mu \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \langle \beta^t [Y_{[p]}, X_{[d]}], \epsilon_{\mu} [Y_{[p]}], \nu [X_{[d]}] \rangle_{[X_{[d]}]} \\ &= \langle \beta^t [Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}} [Y_{[p]}], \nu [X_{[d]}] \rangle_{[X_{[d]}]} \end{aligned} \quad \square$$

We now investigate the representation of face measures by Computation-Activation Networks.

Definition 4. Let $\mathcal{G}$ be a hypergraph which nodes include $[p]$. We say that a face $\mathcal{F}$ is representable by $\mathcal{G}$ if and only if there is a set $\mathcal{U}$ of basis vectors with $\mathcal{U} \cap \mathcal{N}_{t,\nu} = \mathcal{N}_{t,\nu}^{\mathcal{F}}$ and there is a tensor network $\kappa^{\mathcal{G}}$ with respect to the hypergraph $\mathcal{G}$ such that

$$\langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]} = \sum_{v \in \mathcal{U}} \epsilon_v [Y_{[p]}] .$$

We call any such tensor network a face activating tensor network.

Lemma 4. If any only if a face is representable by a hypergraph $\mathcal{G}$ we have

$$\frac{1}{\langle \nu^{t,\mathcal{F}} [X_{[d]}] \rangle_{[\emptyset]}} \cdot \nu^{t,\mathcal{F}} [X_{[d]}] \in \Lambda^{t,\mathcal{G},\nu} .$$

Proof. For any $\mathcal{U}$ with $\mathcal{U} : \mathcal{U} \cap \mathcal{N}_{t,\nu} = \mathcal{N}_{t,\nu}^{\mathcal{F}}$ and tensor network $\kappa^{\mathcal{G}}$ respecting the assumptions of Def. 4 we have

$$\begin{aligned} \left\langle \langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]}, \beta^t [Y_{[p]}, X_{[d]}] \right\rangle_{[X_{[d]}]} &= \left\langle \sum_{v \in \mathcal{U}} \epsilon_v [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \right\rangle_{[X_{[d]}]} \\ &= \left\langle \sum_{v \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_v [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \right\rangle_{[X_{[d]}]} . \end{aligned}$$

Here we used in the second equation that only the vertices in $\mathcal{N}_{t,\nu}$ are in the image of t . It follows, that

$$\frac{1}{\langle \nu^{t,\mathcal{F}} [X_{[d]}] \rangle_{[\emptyset]}} \cdot \nu^{t,\mathcal{F}} [X_{[d]}] = \frac{\left\langle \langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]}, \beta^t [Y_{[p]}, X_{[d]}] \right\rangle_{[X_{[d]}]}}{\left\langle \langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]}, \beta^t [Y_{[p]}, X_{[d]}], \nu [X_{[d]}] \right\rangle_{[\emptyset]}}$$

and thus $\nu^{t,\mathcal{F}} [X_{[d]}] \in \Lambda^{t,\mathcal{G},\nu}$. □

Let us now investigate, which normalized face measures can be computed using t and a hypergraph $\mathcal{G}$.

Example 1 (Vertices). Vertices $\mathcal{F}$ are proper faces of affine dimension 0, that is they consist in single vectors. Since all vertices are in the image $\sigma^t(\mathcal{X})$, there exists an index tuple $x_{[d]} \in \mathcal{X}$ such that $\nu [X_{[d]} = x_{[d]}] = 1$ and

$$\mathcal{F} = \{\sigma^t [X_{[d]} = x_{[d]}, L]\} .$$

Then $\kappa^{\mathcal{F}} [Y_{[p]}]$ is the one-hot encoding of the by an interpretation map I assigned index to $\sigma^t [X_{[d]} = x_{[d]}, L]$, that is

$$\kappa^{\mathcal{F}} [Y_{[p]}] = \epsilon_{I^{-1}(\sigma^t [X_{[d]} = x_{[d]}, L])} [Y_{[p]}] .$$

In particular, the activation core is elementary and the face measure to any vertex is in $\Lambda^{t,\text{EL},\nu}$.

While vertices are the minimal non-vanishing faces in the face-lattice (see Ziegler (2013)), we now show that also the maximal face, namely the polytope itself, is representable with respect to the elementary hypergraph EL.

Example 2 (Maximal face). The maximal face $\mathcal{F}^{\emptyset} = \mathcal{M}_{t,\nu}$ coincides with the mean polytope itself and is given by the choice $\theta_{\emptyset} [L] = 0 [L]$. In this case the corresponding activation tensor to the face measure is trivial, that is

$$\kappa^{\emptyset} [Y_{[p]}] = \mathbb{I} [Y_{[p]}] .$$

$\kappa^{\emptyset}$ is elementary and the normalized face measure $\nu^{t,\emptyset}$ to the maximal face is in $\Lambda^{t,\text{EL}}$.

Extending Example 1, we can provide a coarse estimation of the hypergraph $\mathcal{G}$ required to decompose $\kappa^{\mathcal{F}}$ for generic faces $\mathcal{F}$.

Lemma 5. Any face $\mathcal{F}$ is representable by a CP graph with hidden rank

$$r = \min (|\mathcal{N}_{t,\nu}^{\mathcal{F}}|, |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| + 1) .$$

Proof. We show the claim by constructing two face activating tensor networks to $\mathcal{F}$ in a CP hypergraph with hidden rank $|\mathcal{N}_{t,\nu}^{\mathcal{F}}|$ and in a CP hypergraph with hidden rank $|\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| + 1$. To show the first representation we enumerate the vertices by a variable I with dimension $r = |\mathcal{N}_{t,\nu}^{\mathcal{F}}|$, i.e. $\mathcal{N}_{t,\nu}^{\mathcal{F}} = \{v^i[L] : i \in [n]\}$. Then we define for $\ell \in [p]$ core tensors $\tau^\ell[Y_\ell, I]$

$$\tau^\ell[Y_\ell, I = i] = \epsilon_{v^i[L=\ell]}[Y_\ell] .$$

Then we have

$$\langle \{\tau^\ell[Y_\ell, I] : \ell \in [p]\} \rangle_{[Y_{[p]}]} = \sum_{v \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_v[Y_{[p]}]$$

and thus have found an activation tensor network in a CP graph with hidden rank $|\mathcal{N}_{t,\nu}^{\mathcal{F}}|$ representing the face $\mathcal{F}$.

We continue with the second representation, for which we enumerate the set $\mathcal{N}_{t,\nu}/\mathcal{N}_{t,\nu}^{\mathcal{F}}$ by $v^i[L]$ where $i \in [|\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}|]$. We define variable I with dimension $r = |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| + 1$ and define for $\ell \in [p]$ core tensors

$$\tau^\ell[Y_\ell, I = i] = \begin{cases} -\epsilon_{v^i[L=\ell]}[Y_\ell] & \text{if } i < |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| \\ \mathbb{I}[Y_\ell] & \text{if } i = |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| \end{cases} .$$

We then have

$$\begin{aligned} \langle \{\tau^\ell[Y_\ell, I] : \ell \in [p]\} \rangle_{[Y_{[p]}]} &= \mathbb{I}[Y_{[p]}] - \sum_{v \in \mathcal{N}_{t,\nu}/\mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_v[Y_{[p]}] \\ &= \sum_{v \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_v[Y_{[p]}] + \sum_{v \in (\times_{\ell \in [p]} [p_\ell]) / \mathcal{N}_{t,\nu}} \epsilon_v[Y_{[p]}] . \end{aligned}$$

We have thus found a face activating tensor network for $\mathcal{F}$ in a CP format with hidden rank $r = |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| + 1$. $\square$

2.3 Examples of polytopes with elementary representable faces

We first show in the following example, that all faces of a hypercube are representable by elementary activation tensors.

Example 3 (Hypercube). *In cases where $X_{[d]}$ are boolean and we have $\ell \in [p]$ features*

$$f_\ell[X_{[d]} = x_{[d]}] = x_\ell$$

the mean polytope is the hypercube

$$\mathcal{M}_{\{f_\ell : \ell \in [p]\}, \mathbb{I}} = [0, 1]^p .$$

The non-empty faces in the face lattice $L([0, 1]^p)$ can be enumerated by subsets $A \subset [p]$ and indices $y_A \in \times_{\ell \in A} [2]$ and represented by the cartesian products

$$\mathcal{F}^{(A, y_A)} = \times_{\ell \in [p]} \mathcal{I}^{\ell, (A, y_A)}$$

where

$$\mathcal{I}^{\ell, (A, y_A)} = \begin{cases} [0, 1] & \text{if } \ell \notin A \\ \{y_\ell\} & \text{if } \ell \in A \end{cases} .$$

Each of these faces can be represented with respect to the elementary graph EL, namely by the tensor product of leg vectors

$$\kappa^{\ell, (\mathcal{U}, y_\mathcal{U})}[Y_\ell] = \begin{cases} \mathbb{I}[Y_\ell] & \text{if } \ell \notin A \\ \epsilon_{y_\ell}[Y_\ell] & \text{if } \ell \in A \end{cases} .$$

Example 4 (Simplices). *The $(p-1)$ -dimensional standard simplex $\mathcal{M}_{\Delta, p-1}$ in $[0, 1]^p$ is the convex hull of the vertex sets*

$$\mathcal{N} = \{\epsilon_\ell[L] : \ell \in [p]\} .$$

The simplex is the mean polytope of the p -dimensional statistic with

$$t_\ell : [p] \rightarrow [2] \quad , \quad t_\ell [\tilde{\ell}] = \begin{cases} 1 & \text{if } \tilde{\ell} = \ell \\ 0 & \text{if } \tilde{\ell} \neq \ell \end{cases}.$$

This is sometimes referred to as the probability simplex, since it represents all distributions of a discrete variable with p states.

The face lattice of the simplex consists is enumerated by subsets of $[p]$ as

$$L(\mathcal{M}_{\Delta, p-1}) = \{\mathcal{F}^A : A \subset [p]\}$$

where the faces are

$$\mathcal{F}^A = \text{conv}(\epsilon_\ell[L] : \ell \in A)$$

and therefore itself $(|A|-1)$ -dimensional standard simplices. The partial order of the faces coincides with the inclusion order of subsets A .

2.4 Partition into Relative Interiors of Faces

Let us now introduce relative interiors, which enables us to find disjoint partitions of the mean polytope.

Definition 5 (Relative Interior). *The relative interior $(\mathcal{U})^\circ$ of an arbitrary set $\mathcal{U} \subset \mathbb{R}^p$ is the interior of $\mathcal{U}$ in the affine subspace $\text{conv}(\mathcal{U})$.*

Lemma 6. *Any polytope is a disjoint union of the relative interiors of its faces, that is*

$$\mathcal{M}_{t,\nu} = \bigcup_{\mathcal{F} \in L(\mathcal{M}_{t,\nu})} (\mathcal{F})^\circ.$$

Proof. For any $\mu \in \mathcal{M}_{t,\nu}$ we find a face such that $\mu \in \mathcal{F}$. If $\mu \notin (\mathcal{F})^\circ$, then there is a face $\mathcal{F}^{\tilde{\mathcal{F}}} \subset \mathcal{F}$ of smaller affine dimension such that $\mu \in \mathcal{F}^{\tilde{\mathcal{F}}}$. When continuing this process we reach a face such that $\mu \in (\mathcal{F})^\circ$, since the faces with affine dimension 0 are vertices and they coincide with their relative interior because they contain a single vector. $\square$

Definition 6. *To each $\mu \in \mathcal{M}_{t,\nu}$ we denote the unique face $\mathcal{F}$ with $\mu \in (\mathcal{F})^\circ$ by $\mathcal{F}(\mu)$.*

3 Exponential Families

It is well known that exponential families are specific maximum entropy distributions. However, not any maximum entropy distribution is an exponential family member. To prepare for the characterization of generic maximum entropy distributions, we will in this section represent exponential family members as Computation-Activation Networks. We further show that a maximum entropy distribution is a exponential family member, if and only if its mean parameter is in the relative interior of the mean polytope.

3.1 Exponential Families

Definition 7 (Exponential Family). *Given a statistic function*

$$t : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}^p$$

and a base measure

$$\nu : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}_0^+$$

with $\langle \nu \rangle_{[\emptyset]} \neq 0$, the set $\Gamma^{t,\nu} = \{\mathbb{P}^{(t,\theta,\nu)} : \theta[L] \in \mathbb{R}^p\} \subset \Lambda^\nu$ of probability distributions

$$\mathbb{P}^{(t,\theta,\nu)}[X_{[d]}] = \frac{\exp \left[\langle \sigma^t[X_{[d]}, L], \theta[L] \rangle_{[X_{[d]}]} \right]}{\left\langle \exp \left[\langle \sigma^t[X_{[d]}, L], \theta[L] \rangle_{[X_{[d]}]} \right], \nu[X_{[d]}] \right\rangle_{[\emptyset]}}$$

is called the exponential family to t .

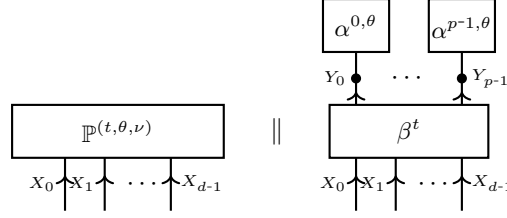


Figure 2: Representation of a member in an exponential family by a Computation-Activation Network with elementary activation. Since the right hand side is not normalized both sides are equal up to a constant.

To present a tensor network representation, we introduce image interpretation maps

$$I_\ell : [|\text{im}(t_\ell)|] \rightarrow \text{im}(t_\ell) ,$$

which enumerate the possible values of each feature. We treat these maps as tensors with in a variable Y_ℓ with values in $[|\text{im}(t_\ell)|]$.

3.2 Representation as elementary Computation-Activation Networks

Theorem 2 (Exponential Families are elementary Computation-Activation Networks). *Given any base measure ν and a sufficient statistic t we enumerate for each coordinate $\ell \in [p]$ the image $\text{im}(t_\ell)$ by a variable Y_ℓ taking values in $[|\text{im}(t_\ell)|]$, given an interpretation map*

$$I_\ell : [|\text{im}(t_\ell)|] \rightarrow \text{im}(t_\ell) .$$

For any canonical parameter vector $\theta[L] \in \mathbb{R}^p$ we build the activation cores $\alpha^{\ell, \theta}[Y_\ell]$ for each coordinate $y_\ell \in [|\text{im}(t_\ell)|]$ by

$$\alpha^{\ell, \theta}[Y_\ell = y_\ell] = \exp[\theta[L = \ell] \cdot I_\ell(y_\ell)]$$

and have (see Figure 2)

$$\mathbb{P}^{(t, \theta, \nu)}[X_{[d]}] = \frac{\langle \{\beta^{t_\ell}[Y_\ell, X_{[d]}] : \ell \in [p]\} \cup \{\alpha^{\ell, \theta}[Y_\ell] : \ell \in [p]\} \rangle_{[X_{[d]}]}}{\langle \{\beta^{t_\ell}[Y_\ell, X_{[d]}] : \ell \in [p]\} \cup \{\alpha^{\ell, \theta}[Y_\ell] : \ell \in [p]\} \cup \{\nu[X_{[d]}]\} \rangle_{[\emptyset]}} .$$

Proof. For each $x_{[d]} \in \times_{k \in [d]} [m_k]$ we have

$$\begin{aligned} & \langle \{\beta^{t_\ell}[Y_\ell, X_{[d]}] : \ell \in [p]\} \cup \{\alpha^{\ell, \theta}[Y_\ell] : \ell \in [p]\} \rangle_{[X_{[d]} = x_{[d]}]} \\ &= \prod_{\ell \in [p]} \langle \beta^{t_\ell}[Y_\ell, X_{[d]} = x_{[d]}], \alpha^{\ell, \theta}[Y_\ell] \rangle_{[\emptyset]} \\ &= \prod_{\ell \in [p]} \exp[\theta[L = \ell] \cdot I_\ell(t_\ell[x_{[d]}])] \\ &= \exp \left[\sum_{\ell \in [p]} \theta[L = \ell] \cdot I_\ell(t_\ell[x_{[d]}]) \right] \\ &= \exp \left[\langle \sigma^t[X_{[d]}, L], \theta[L] \rangle_{[X_{[d]} = x_{[d]}]} \right] . \end{aligned}$$

Therefore we have

$$\langle \{\beta^{t_\ell}[Y_\ell, X_{[d]}] : \ell \in [p]\} \cup \{\alpha^{\ell, \theta}[Y_\ell] : \ell \in [p]\} \rangle_{[X_{[d]}]} = \exp \left[\langle \sigma^t[X_{[d]}, L], \theta[L] \rangle_{[X_{[d]}]} \right] .$$

The claim follows, since this implies that also the contraction of both sides with $\nu[X_{[d]}]$ is equivalent. $\square$

We will use the following well known property (see e.g. Brown (1987)), that there is a one-to-one map between the canonical parameters and the mean parameters.

Lemma 7. *The set of mean parameters of the members of an exponential family is the relative interior of the mean polytope. For $\theta, \tilde{\theta} \in \mathbb{R}^p$ with*

$$\langle \mathbb{P}^{t,\theta,\nu}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]} = \langle \mathbb{P}^{t,\tilde{\theta},\nu}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]}$$

we furthermore have $\mathbb{P}^{t,\theta,\nu}[X_{[d]}] = \mathbb{P}^{t,\tilde{\theta},\nu}[X_{[d]}]$.

Proof. See The 3.3 in Wainwright and Jordan (2008). □

Based on this property we define the forward and backward mappings to an exponential family.

Definition 8. *The forward map of an exponential family is the map*

$$F^{(t,\nu)} : \mathbb{R}^p \rightarrow (\mathcal{M}_{t,\nu})^\circ$$

defined as

$$\forall \ell \in [p] : F^{(t,\nu)}(\theta)[L = \ell] = \langle \mathbb{P}^{t,\theta,\nu}[X_{[d]}], t_\ell[X_{[d]}], \nu[X_{[d]}] \rangle_{[\emptyset]}$$

Any map $B^{(t,\nu)} : (\mathcal{M}_{t,\nu})^\circ \rightarrow \mathbb{R}^p$ with $B^{(t,\nu)} \circ F^{(t,\nu)} = \text{Id}$ is called a backward map.

3.3 Maximum entropy distributions

We now show that exponential families are maximum entropy distributions.

Lemma 8. *Any member of an exponential family $\Gamma^{t,\nu}$ is a maximum entropy distribution with respect to the moment constraint on t and the base measure ν .*

Proof. Let $\mathbb{P}^{(t,\theta,\nu)}$ be a exponential family member with canonical parameter θ and let $\mu[L]$ be the corresponding mean parameter. Let $\tilde{\mathbb{P}}[X_{[d]}]$ be any feasible distribution for the maximum entropy problem with respect to the moment constraint on t by $\mu[L]$ and the base measure ν . We also have $\langle \tilde{\mathbb{P}}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]} = \mu[L]$ and thus

$$\begin{aligned} \mathbb{H}_\nu[\tilde{\mathbb{P}}, \mathbb{P}^{(t,\theta,\nu)}] &= - \langle \tilde{\mathbb{P}}[X_{[d]}], \ln[\mathbb{P}^{(t,\theta,\nu)}[X_{[d]}]], \nu[X_{[d]}] \rangle_{[\emptyset]} \\ &= - \langle \tilde{\mathbb{P}}[X_{[d]}], \sigma^t[X_{[d]}, L], \theta[L], \nu[X_{[d]}] \rangle_{[\emptyset]} + A^{(t,\nu)}(\theta) \\ &= - \langle \theta[L], \mu[L] \rangle_{[\emptyset]} + A^{(t,\nu)}(\theta) \\ &= \mathbb{H}_\nu[\mathbb{P}^{(t,\theta,\nu)}] . \end{aligned}$$

With the Gibbs inequality we have if $\tilde{\mathbb{P}} \neq \mathbb{P}^{(t,\theta,\nu)}$

$$\mathbb{H}_\nu[\mathbb{P}^{(t,\theta,\nu)}] - \mathbb{H}_\nu[\tilde{\mathbb{P}}] = \mathbb{H}_\nu[\tilde{\mathbb{P}}, \mathbb{P}^{(t,\theta,\nu)}] - \mathbb{H}_\nu[\tilde{\mathbb{P}}] > 0$$

and thus $\mathbb{H}_\nu[\tilde{\mathbb{P}}] < \mathbb{H}_\nu[\mathbb{P}^{(t,\theta,\nu)}]$. Therefore, any feasible $\tilde{\mathbb{P}}$ different from $\mathbb{P}^{(t,\theta,\nu)}$ have less entropy than $\mathbb{P}^{(t,\theta,\nu)}$ and $\mathbb{P}^{(t,\theta,\nu)}$ is a maximum entropy distribution. □

Together with Lem. 7 stating the existence of an exponential family member for any mean parameter in the relative interior of the mean polytope we arrive at the following theorem.

Theorem 3. *If and only if $\mu[L]$ is in the relative interior of $\mathcal{M}_{t,\nu}$, then the unique solution of the maximum entropy problem is the distribution*

$$\mathbb{P}^{t,\theta,\nu}[X_{[d]}] \in \Gamma^{t,\nu}$$

where for some backward map $B^{t,\nu}$

$$\theta = B^{t,\nu}(\mu) .$$

Proof. By Lem. 7 if

$$\mu[L] \in (\mathcal{M}_{t,\nu})^\circ,$$

there is at least one weight parameter $\theta[L] \in \mathbb{R}^p$ with

$$\langle \mathbb{P}^{t,\theta,\nu}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]} = \mu[L]$$

and $B^{t,\nu}(\mu)$ with respect to any backward map $B^{t,\nu}$ is such one. With Lem. 8 we conclude that $\mathbb{P}^{(t, B^{t,\nu}(\mu), \nu)}$ is the maximum entropy distribution with respect to the moment constraint on t by $\mu[L]$ and the base measure ν . $\square$

Note that while the maximum entropy distribution is unique, the weight vector θ representing the maximum entropy distribution is not always unique. This is reflected by multiple possible choices of backward mappings. Only if the exponential family is minimal (see Wainwright and Jordan (2008)) then the weight vector is unique. However, we did not restrict to such situations, since exponential families on refined base measures are not minimal.

4 Generic maximum entropy distributions

In the previous section we have shown that the members of an exponential family are maximum entropy distribution. However, their mean parameters build the relative interior of the mean polytope, but are never on faces of smaller dimension. We here characterize the maximum entropy distributions with respect to mean parameters on the full mean polytope.

4.1 Main result

Our main result generalizes the maximum entropy characterization of Thm. 3 to arbitrary mean parameters.

Theorem 4 (Generic characterization of Maximum Entropy Solutions). *Let t be a statistic and ν a base measure. For any $\theta[L]$ and $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ the distribution*

$$\mathbb{P}^{(t,\theta,\nu^{t,\mathcal{F}})}$$

is a maximum entropy distribution.

For any $\mu[L]$ the maximum entropy problem has a feasible distribution, if and only if $\mu[L] \in \mathcal{M}_{t,\nu}$. If $\mu[L] \in \mathcal{M}_{t,\nu}$ the maximum entropy distribution is $\mathbb{P}^{(t,\theta,\nu^{t,\mathcal{F}})}$ where $\mathcal{F} = \mathcal{F}(\mu)$ and

$$\theta = B^{t,\nu^{t,\mathcal{F}}}(\mu)$$

where $\nu^{t,\mathcal{F}}$ is the refined base measure (see Def. 3) and $B^{t,\nu^{t,\mathcal{F}}}$ is any backward map to the exponential family $\Gamma^{t,\nu^{t,\mathcal{F}}}$ (see Def. 8).

For any hypergraph $\mathcal{G}$ and any face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ we have that all maximum entropy distributions to mean parameters on $\mathcal{F}^\circ$ are in $\Lambda^{t,\mathcal{G},\nu}$ if and only if $\mathcal{F}$ is representable with respect to $\mathcal{G}$ (see Def. 4)

To prepare for the proof of this theorem we first show in an auxiliary lemma that we can reduce the set of feasible distributions in Problem $P_{t,\mu,\nu}$.

Lemma 9. *For any $\mu[L] \in \mathcal{M}_{t,\nu}$ and a face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ with $\mu[L] \in (\mathcal{F})^\circ$ we have that the solutions of $P_{t,\mu,\nu}$ and $P_{t,\mu,\nu^{t,\mathcal{F}}}$ coincide.*

Proof. By Lem. 3 all feasible distributions are representable by the with the face measure refined base measure. We have that any for $P_{t,\mu,\nu}$ feasible distribution $\mathbb{P}[X_{[d]}]$ satisfies

$$\langle \mathbb{P}[X_{[d]}], \nu^{t,\mathcal{F}} \rangle_{[\emptyset]} = 1$$

and thus $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}$. Conversely, any $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}$ satisfies

$$\langle \mathbb{P}[X_{[d]}], \nu^{t,\mathcal{F}} \rangle_{[\emptyset]} = \langle \mathbb{P}[X_{[d]}], \nu \rangle_{[\emptyset]} = 1$$

and thus $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu}$. Problem $P_{t,\mu,\nu}$ is thus equal to

$$\operatorname{argmax}_{\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}} \mathbb{H}_\nu[\mathbb{P}[X_{[d]}]] \quad \text{subject to} \quad \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]} = \mu^*[L]$$

We further have that any $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}$

$$\mathbb{H}_\nu[\mathbb{P}[X_{[d]}]] = \mathbb{H}_{\nu^{t,\mathcal{F}}}[\mathbb{P}[X_{[d]}]]$$

and arrive together with the above equivalence at the claim. $\square$

Proof of Thm. 4. To show that for any $\theta[L]$ and $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ the distribution $\mathbb{P}^{(t,\theta,\nu^t,\mathcal{F})}$ is a maximum entropy distribution, we apply Lem. 9 and Thm. 3.

If $\mu[L] \notin \mathcal{M}_{t,\nu}$ then Problem $P_{t,\mu,\nu}$ is not feasible by definition. In case $\mu[L] \in \mathcal{M}_{t,\nu}$ we find $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ with $\mathcal{F} = \mathcal{F}(\mu)$ and by Lem. 9 the solution of maximum entropy problem Problem $P_{t,\mu,\nu}$ coincides with $P_{t,\mu,\nu^t,\mathcal{F}}$. Since by Lem. 2 the face is the polytope

$$\mathcal{F} = \mathcal{M}_{t,\nu^t,\mathcal{F}}$$

we find a weight parameter

$$\theta = B^{t,\nu^t,\mathcal{F}}(\mu).$$

and by Thm. 3 $\mathbb{P}^{(t,\theta,\nu^t,\mathcal{F})}$ is the corresponding maximum entropy distribution.

To show the last claim, it is enough to show that for any by $\mathcal{G}$ representable face $\mathcal{F}$ and for any $\theta \in \mathbb{R}^p$ we have

$$\mathbb{P}^{(t,\theta,\nu^t,\mathcal{F})} \in \Lambda^{t,\mathcal{G},\nu}.$$

To this end, we construct activation cores as follows. By Def. 4 and Lem. 4 there is a tensor network $\kappa^{\mathcal{G}}$ such that

$$\nu^{t,\mathcal{F}}[X_{[d]}] = \langle \kappa^{\mathcal{G}} \cup \{\nu[X_{[d]}]\} \rangle_{[X_{[d]}]}.$$

For each $\ell \in [p]$ we find an edge $e(\ell) \in \mathcal{E}$ with $\ell \in e(\ell)$. Now we define a tensor network $\tau^{\mathcal{G}}$ by the cores to $e \in \mathcal{E}$ as

$$\tau^e[X_e] = \langle \{ \kappa^e[X_e] \} \cup \{ \alpha^{\ell,\theta}[Y_\ell] : e = e(\ell) \} \rangle_{[X_e]}.$$

Then, we have

$$\langle \beta^t[Y_{[p]}, X_{[d]}], \langle \tau^{\mathcal{G}} \rangle_{[Y_{[p]}]} \rangle_{[X_{[d]}]} = \langle \beta^t[Y_{[p]}, X_{[d]}], \alpha^\theta[Y_{[p]}], \langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]} \rangle_{[X_{[d]}]}$$

and therefore

$$\mathbb{P}^{(t,\theta,\nu^t,\mathcal{F})} = \frac{\langle \{ \beta^t[Y_{[p]}, X_{[d]}] \} \cup \tau^{\mathcal{G}} \rangle_{[X_{[d]}]}}{\langle \{ \beta^t[Y_{[p]}, X_{[d]}], \nu[X_{[d]}] \} \cup \tau^{\mathcal{G}} \rangle_{[\emptyset]}}.$$

We conclude $\mathbb{P}^{(t,\theta,\nu^t,\mathcal{F})} \in \Lambda^{t,\mathcal{G},\nu}$.

□

4.2 Family of maximum entropy distributions

The family of maximum entropy distributions is the set

$$\Gamma^{t,\nu} = \{ \mathbb{P}[X_{[d]}] : \exists \mu \in \mathcal{M}_{t,\nu} : \mathbb{P}[X_{[d]}] \text{ is a solution of Problem } P_{t,\mu,\nu} \}.$$

Thm. 4 suggest a parametrization of this family as tuples of corresponding faces $\mathcal{F}$ and weight vectors $\mu[L]$. We thus define the canonical parameter space

$$\mathcal{P}_{t,\nu} := L(\mathcal{M}_{t,\nu}) \times \mathbb{R}^p$$

and have

$$\Gamma^{t,\nu} = \left\{ \frac{\langle \alpha^\theta[Y_{[p]}], \kappa^{\mathcal{F}}[Y_{[p]}], \beta^t[Y_{[p]}, X_{[d]}] \rangle_{[X_{[d]}]}}{\langle \alpha^\theta[Y_{[p]}], \kappa^{\mathcal{F}}[Y_{[p]}], \beta^t[Y_{[p]}, X_{[d]}], \nu[X_{[d]}] \rangle_{[\emptyset]}} : \theta[L] \in \mathbb{R}^p, \mathcal{F} \in \mathcal{P}_{t,\nu} \right\}.$$

Note that these families are disjoint, since each member of an exponential family has support by the support of the face measure.

5 Characterization for boolean statistics

We here study the face CP ranks in case of boolean statistics. We further show that any elementary Computation-Activation Network to boolean statistics is a maximum entropy distribution.

For boolean statistics $t : \times_{k \in [d]} [m_k] \rightarrow \times_{\ell \in [p]} [2]$ the mean polytope is a subset of the cube $[0, 1]^p$. In this case, any boolean vector in $\mathcal{M}_{t,\nu}$ is a vertex. It follows, that any distribution reproducing a mean parameter $\mu[L]$ on the relative interior of $\mathcal{M}_{t,\nu}$ is positive with respect to ν .

We apply the exponential distribution characterization of the maximum entropy distribution and get that the maximum entropy distribution is in $\Lambda^{t,\text{EL}}$, if and only if the face measure is in $\Lambda^{t,\text{EL}}$. This is exactly the case, when the face is an intersection of the mean polytope with a face of the cupe $[0, 1]^p$.

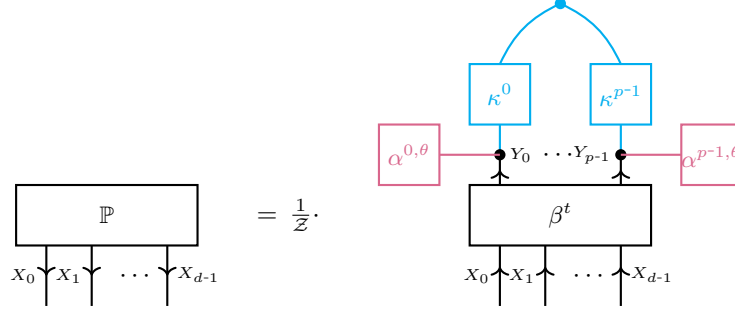


Figure 3: Tensor network decomposition of maximum entropy distributions to the constraint $\mu[L] = \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L] \rangle_{[L]}$. Blue: Constraint activation cores κ^ℓ in a CP decomposition, representing the face measure to the minimal face, such that $\mu \in \mathcal{F}^\mathcal{F}$. Red: Probabilistic activation cores $\alpha^{\ell, \theta}[Y_\ell]$ in an elementary decomposition, where each leg core is a scaled exponentials evaluated on the enumerated image $\text{im}(t_\ell)$.

5.1 Characterization of elementary representable faces

Orienting on Example 3 we define cube-likeness of faces and polytopes.

Definition 9. We say that a face $\mathcal{F}$ of $\mathcal{M}_{t, \mathbb{I}}$ is cube-like, if it is empty or there is $A \subset [p]$ and $y_A \in \times_{\ell \in A} [2]$ such that

$$\mathcal{F} = \mathcal{M}_{t, \nu} \cap \mathcal{F}^{(A, y_A)}.$$

We further say that a polytope $\mathcal{M}_{t, \mathbb{I}}$ is cube-like, if all faces $\mathcal{F} \in L(\mathcal{M}_{t, \mathbb{I}})$ are cube-like.

We now show that a face is cube-like if and only if it is representable by an elementary tensor.

Theorem 5. Let t be a boolean statistic, ν a base measure and $\mathcal{F}$ be a face of $\mathcal{M}_{t, \mathbb{I}}$. Then the following are equivalent:

- (i) $\mathcal{F}$ is cube-like (see Def. 9).
- (ii) $\mathcal{F}$ is representable by an elementary tensor (see Def. 4).

Proof. If the face is empty, i.e. $\mathcal{F} = \emptyset$, it is by definition cube-like and has $0[Y_{[p]}]$ as an elementary activation tensor. We therefore assume in the following $\mathcal{F} \neq \emptyset$.

(i) $\Rightarrow$ (ii): Let us assume that $\mathcal{F}$ is cube-like, that is there is $A \subset [p]$ and $y_A \in \times_{\ell \in A} [2]$ such that $\mathcal{F} = \mathcal{M}_{t, \nu} \cap \mathcal{F}^{(A, y_A)}$. We use that $\mathcal{F}$, $\mathcal{M}_{t, \nu}$ and $\mathcal{F}^{(A, y_A)}$ are the convex hulls of the cube vertex sets $\mathcal{N}_{t, \mathcal{M}}^\mathcal{F}$, $\mathcal{N}_{t, \mathcal{M}}$ and

$$\mathcal{N}^{(A, y_A)} = \{v[L] : \forall \ell \in A v[L = \ell] = y_A\}, \quad (1)$$

which implies that

$$\mathcal{N}_{t, \mathcal{M}}^\mathcal{F} = \mathcal{N}^{(A, y_A)} \cap \mathcal{N}_{t, \mathcal{M}}.$$

Thus, we can choose $\mathcal{U} = \mathcal{N}_{t, \mathcal{M}}^\mathcal{F}$ for the representation of the face $\mathcal{F}$ (see Def. 4) and further have that

$$\sum_{v[L] \in \mathcal{N}^{(A, y_A)}} = \bigotimes_{\ell \in [p]} \kappa^{\ell, (\mathcal{U}, y_{\mathcal{U}})}[Y_\ell].$$

We have thus found an elementary activation tensor for the face $\mathcal{F}$.

(ii) $\rightarrow$ (i) Conversely, let $\xi[Y_{[p]}]$ be an elementary activation tensor of the face $\mathcal{F}$ in $\mathcal{M}_{t, \mathbb{I}}$. Since $\xi[Y_{[p]}]$ is the sum of different one-hot encodings it is boolean and we find an elementary decomposition $\xi[Y_{[p]}] = \bigotimes_{\ell \in [p]} \xi^\ell[Y_\ell]$ such that the leg vectors $\xi[Y_{[p]}]$ are boolean. Since $\mathcal{F} \neq \emptyset$ we further have for $\ell \in [p]$ that $\xi^\ell[Y_\ell] \neq 0[Y_\ell]$, and thus $\xi^\ell[Y_\ell] \in \{\epsilon_0[Y_\ell], \epsilon_1[Y_\ell], \mathbb{I}[Y_\ell]\}$. We construct a set $A \subset [p]$

$$A = \{\ell : \xi^\ell[Y_\ell] \neq \mathbb{I}[Y_\ell]\}$$

and a tuple

$$y_\ell = \begin{cases} 1 & \text{if } \xi^\ell[Y_\ell] = \epsilon_1[Y_\ell] \\ 0 & \text{if } \xi^\ell[Y_\ell] = \epsilon_0[Y_\ell] \end{cases}.$$

By construction $\xi^\ell[Y_\ell] = \kappa^{\ell,(\mathcal{U},y_\ell)}[Y_\ell]$ (see Example 3) follows for all $\ell \in [p]$ and therefore $\xi[Y_{[p]}] = \kappa^{(\mathcal{U},y_\ell)}[Y_{[p]}]$. We therefore have for the set (1)

$$\xi[Y_{[p]}] = \sum_{v \in \mathcal{N}^{A,y_A}} \epsilon_v[Y_{[p]}]$$

and since $\xi[Y_{[p]}]$ is an activation tensor for $\mathcal{F}$, that

$$\mathcal{N}_{t,\mathcal{M}}^{\mathcal{F}} = \mathcal{N}^{A,y_A} \cap \mathcal{N}_{t,\mathcal{M}}.$$

Taking convex hulls on both sides, this is equivalent to

$$\mathcal{F} = \mathcal{F}^{(A,y_A)} \cap \mathcal{M}_{t,\nu}.$$

We conclude that $\mathcal{F}$ is cube-like. $\square$

Thm. 5 implies that hypercubes (see Example 3) and simplices (see Example 4) are cube-like polytopes, since their faces are intersections with faces of the hypercube.

5.2 Set of maximum entropy distributions

Theorem 6. *Any distribution in $\Lambda^{t,\text{EL},\nu}$ is a maximum entropy distribution with respect to (t, μ, ν) where μ is its mean parameter. Any maximum entropy distribution is realized by $\Lambda^{t,\text{EL},\nu}$ if and only if the mean parameter is in the relative interior of a cube-like face.*

Proof. First claim by decomposing any elementary tensor into exponential and hard activation core. Second claim by characterization of elementary faces by cube-likeness. $\square$

We can now use the same notation as applied for hypercubes to classify the faces of a cube-like polytope.

5.3 Interpretation by propositional formulas

We can understand each feature as a propositional formula and the variables $X_{[d]}$ as atoms (possibly after a binarization).

Each vertex of the cube, which is not a vertex of the polytope corresponds with the unsatisfiability of a formula

$$\bigwedge_{\ell \in [p]} \neg^{1-\mu[L=\ell]} f_\ell[X_{[d]}]$$

which is equal with any of the entailment statements for $A \subset [p]$

$$\left(\bigwedge_{\ell \in A} \neg^{1-\mu[L=\ell]} f_\ell[X_{[d]}] \right) \models \left(\bigwedge_{\ell \in A} \neg^{\mu[L=\ell]} f_\ell[X_{[d]}] \right).$$

Along this interpretation we can easily construct examples of statistics, which polytopes are not cube-like.

Example 5 (Maximum entropy distribution with non-elementary activation cores). *Consider two atomic variables X_0 and X_1 and a statistic $\mathcal{F}$ consisting in the formulas*

$$f_0[X_0, X_1] = (X_0 \wedge X_1) \quad , \quad f_1[X_0, X_1] = (X_0 \Rightarrow X_1)$$

with the coordinatewise expressions

$$f_0[X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1] \quad , \quad f_1[X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} [X_0, X_1].$$

We can think of X_0 as a feature on an invoice, and X_1 as a feature on the accounting proposal.

From this we have

$$\begin{aligned} \beta^{(f_0, f_1)} [Y_0 = 0, Y_1 = 0, X_0, X_1] &= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} [X_0, X_1] \quad , \quad \beta^{(f_0, f_1)} [Y_0 = 0, Y_1 = 1, X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} [X_0, X_1] \quad , \\ \beta^{(f_0, f_1)} [Y_0 = 1, Y_1 = 0, X_0, X_1] &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} [X_0, X_1] \quad \text{and} \quad \beta^{(f_0, f_1)} [Y_0 = 1, Y_1 = 1, X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1] . \end{aligned}$$

Since the only vanishing slice of $\beta^{\mathcal{F}}$ with respect to the head variables is that to $y_{0,1} = (1, 0)$, the vertices of the mean polytope are the vectors to the other head indices. The mean polytope is the convex hull of these vertices

$$\mathcal{M}_{(f_0, f_1)} = \text{conv} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} [L], \begin{pmatrix} 0 \\ 1 \end{pmatrix} [L], \begin{pmatrix} 1 \\ 1 \end{pmatrix} [L] \right) .$$

This polytope has a non cube-like face (sketched blue in Figure 4), which is the convex hull of the vertices $[00]^T$, $[11]^T$. This face is parametrized by the (CP-rank 2) hard activation core

$$\kappa^{(0,0),(1,1)} [Y_0, Y_1] = \epsilon_{(0,0)} [Y_0, Y_1] + \epsilon_{(1,1)} [Y_0, Y_1] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} [Y_0, Y_1]$$

and has the face measure

$$\left\langle \kappa^{(0,0),(1,1)} [Y_0, Y_1], \beta^{\mathcal{F}} [Y_0, Y_1, X_0, X_1] \right\rangle_{[X_0, X_1]} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} [X_0, X_1] .$$

Any mean parameter μ on the interior of that face can be parametrized by a scalar $\lambda \in (0, 1)$

$$\mu_\lambda [L] = (\lambda \quad \lambda) [L] .$$

With the canonical parameters $\theta [L] \in \mathbb{R}^2$ of the maximum entropy distributions on this face by

$$\mathbb{P} [X_0, X_1] = \frac{1}{1 + \exp [\theta [L = 0] + \theta [L = 1]]} \begin{pmatrix} 0 & 0 \\ 1 & \exp [\theta [L = 0] + \theta [L = 1]] \end{pmatrix} [X_0, X_1]$$

we get the correspondence by the sigmoid

$$\lambda = \frac{1}{1 + \exp [-(\theta [L = 0] + \theta [L = 1])]} .$$

Note, that the hard activation core $\kappa^{(0,0),(1,1)} [Y_0, Y_1]$ to the blue face is the only non-elementary activation core. While the vertices have always elementary cores, the further non-vertex faces have elementary activation cores

$$\kappa^{(0,0),(1,0),(1,1)} [Y_0, Y_1] = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} [Y_0, Y_1] = \mathbb{I} [Y_0] \otimes \mathbb{I} [Y_1] \quad , \quad \kappa^{(0,0),(1,0)} [Y_0, Y_1] = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} [Y_0, Y_1] = \mathbb{I} [Y_0] \otimes \epsilon_0 [Y_1] \quad ,$$

$$\kappa^{(1,0),(1,1)} [Y_0, Y_1] = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} [Y_0, Y_1] = \epsilon_0 [Y_0] \otimes \mathbb{I} [Y_1] .$$

The maximum entropy distributions to mean parameters on the interior of all other faces than the blue face are represented by Computation-Activation Networks with only elementary activation cores.

Example 6 (Atomic formulas). Let us consider the case of atomic formulas. The mean polytope in this case is the d -dimensional hypercube (see Example 3)

$$\mathcal{M}_{\mathcal{F}[d], \mathbb{I}} = [0, 1]^d$$

which is called a simple polytope, since each vertex is contained in the minimal number of d facets.

The faces of a hypercube are enumerated in the following way. Each face is characterized by the projections onto each variable, which is either $\{0\}$, $\{1\}$ or $[0, 1]$. The projections are represented by the tuple $(\mathcal{U}, y_{\mathcal{U}})$ defined in the following way:

- We define the set $\mathcal{U} \subset [d]$ of variables, such that the projection onto the variable is $\{0\}$ or $\{1\}$
- We define to each $\ell \in \mathcal{U}$ an index $y_\ell = 0$ if the projection is $\{0\}$ and $y_\ell = 1$ if the projection is $\{1\}$.

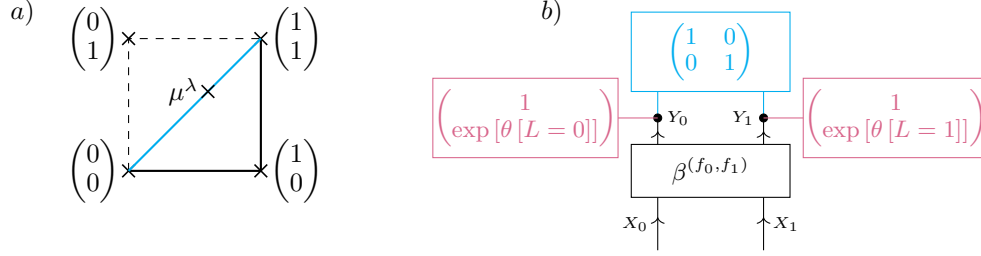


Figure 4: a) Mean polytope of the statistic $\mathcal{F} = (X_0 \wedge X_1, X_0 \Rightarrow X_1)$ (thick), as a subset of the cube $[0, 1]^2$ (dashed). The blue line is the face of the polytope, which is not cube like, that is not an intersection of the polytope with the faces of the polytope. We further define for $\lambda \in (0, 1)$ a mean parameter $\mu_\lambda[L] = [\lambda \lambda]^T$ which is on the interior of the blue face. b) Corresponding Computation-Activation Network being the maximum entropy distribution reproducing $\mu_\lambda[L]$, when λ is the sigmoid of $\theta[L = 0] + \theta[L = 1]$.

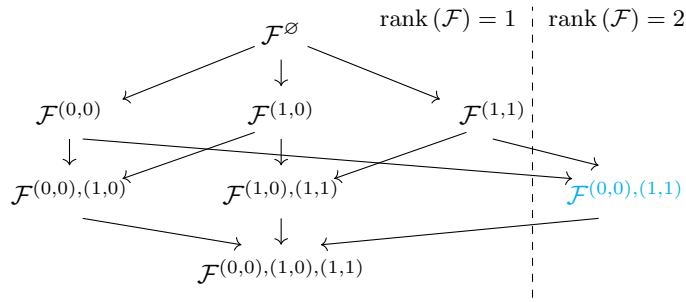


Figure 5: Face lattice $L((X_0 \wedge X_1, X_0 \Rightarrow X_1))$ to Example 5. The directed arrows represent inclusion of the faces, which is a partial order of the faces. The face $\mathcal{F}^{(0,0),(1,1)}$ is the only face, which is not representable by an elementary hypergraph. It is representable in a CP with hidden rank 2

Trivially, each face of the hypercube is a cube face and $\mathcal{M}_{\mathcal{F}_{[d]}}^{\text{EL}} = \mathcal{M}_{\mathcal{F}_{[d]}}$.

More general: If and only if no combination of possibly negated formulas is unsatisfiable, then the mean polytope is a hypercube.

Example 7 (Minterm formulas). The set of minterm formulas is indexed by $x_{[d]} \in \times_{k \in [d]} [m_k]$ and given by

$$f_{x_{[d]}}[X_{[d]}] = \bigwedge_{\ell \in [d]} \neg^{1-x_\ell} f_\ell[X_{[d]}] = \epsilon_{x_{[d]}}[X_{[d]}]$$

where by $f_\ell[X_{[d]}]$ we denote the ℓ -th atomic formula (see Example 6). The mean polytope is in the case of the minterm statistic (also referred to as universal statistic) and a boolean base measure ν the standard simplex of dimension

$$\left| \left\{ x_{[d]} \in \times_{k \in [d]} [m_k] : \nu[X_{[d]} = x_{[d]}] \neq 0 \right\} \right| - 1,$$

that is the set

$$\mathcal{M}_{\mathcal{F}_\wedge, \nu}^{\text{EL}} = \text{conv} \left(\epsilon_{x_{[d]}}[Y_{[p]}] : x_{[d]} \in \times_{k \in [d]} [m_k], \nu[X_{[d]} = x_{[d]}] \neq 0 \right).$$

In this case, $\Lambda^{\mathcal{F}_\wedge, \text{EL}}$ contains any distribution and therefore trivially realizes any mean parameter in $\mathcal{M}_{\mathcal{F}_\wedge, \mathbb{I}}$.

The faces of the standard simplex are itself standard simplices to base measures $\tilde{\nu}$ with $\tilde{\nu} \prec \nu$. We store them by the tuple $(\mathcal{U}, y_\mathcal{U})$, where $\mathcal{U}$ is the support of $\tilde{\nu}$ and $y_\ell = 0$ for $\ell \in \mathcal{U}$. Each face is a cube face, since it is the intersection of $\Lambda^{\mathcal{F}_\wedge, \text{EL}}$ with the cube face $(\mathcal{U}, y_\mathcal{U})$. In particular, we have $\mathcal{M}_{\mathcal{F}_\wedge}^{\text{EL}} = \mathcal{M}_{\mathcal{F}_\wedge}$.

More general the mean polytope is a standard simplex, if and only if each formula contradicts all others.

Example 8 (TT representation of diagonal faces). *Consider the vertex set*

$$\mathcal{N} := \{0, 1\}^p / v[L]$$

convex polytope. We are interested in the face $\mathcal{F}^{\triangleleft, 1}$ with the normal $\frac{1}{2}(2v[L] - \mathbb{I}[L])$. Its vertices are the p elements of $\{0, 1\}^p$, which differ from $v[L]$ in exactly one coordinate (i.e. those with Hamming distance of 1 from $v[L]$). We denote these vertices by v^ℓ for $\ell \in [p]$, which coordinates to $\tilde{\ell} \in [p]$ are

$$v^\ell \left[L = \tilde{\ell} \right] = \begin{cases} v \left[L = \tilde{\ell} \right] & \text{if } \tilde{\ell} \neq \ell \\ 1 - v \left[L = \tilde{\ell} \right] & \text{if } \tilde{\ell} = \ell \end{cases}.$$

We now represent their sum in an TT with hidden ranks $r_0 = \dots = r_{p-2} = 2$. The boolean hidden variables are denoted by $I_{[p-1]}$ and can be interpreted as indicators, whether the coordinate flip has happened in $[\ell]$ coordinates. We now construct a TT cores for $\ell = 0$ by

$$\tau^0 [Y_0, I_0] = \epsilon_{v[L=0]} [Y_0] \otimes \epsilon_0 [I_0] + \epsilon_{1-v[L=0]} [Y_0] \otimes \epsilon_1 [I_0]$$

further for $\ell \notin \{0, p-1\}$ the cores

$$\begin{aligned} \tau^\ell [I_{\ell-1}, Y_\ell, I_\ell] &= \epsilon_0 [I_{\ell-1}] \otimes \epsilon_{1-v[L=\ell]} [Y_\ell] \otimes \epsilon_1 [I_\ell] + \epsilon_0 [I_{\ell-1}] \otimes \epsilon_{v[L=\ell]} [Y_\ell] \otimes \epsilon_0 [I_\ell] \\ &\quad + \epsilon_1 [I_{\ell-1}] \otimes \epsilon_{v[L=\ell]} [Y_\ell] \otimes \epsilon_1 [I_\ell] \end{aligned}$$

and for $\ell = p-1$

$$\tau^{p-1} [I_{p-1}, Y_{p-1}] = \epsilon_0 [I_{p-1}] \otimes \epsilon_{1-v[L=p-1]} [Y_{p-1}] + \epsilon_1 [I_{p-1}] \otimes \epsilon_{v[L=p-1]} [Y_{p-1}].$$

For this tensor network in the TT format we have

$$\sum_{\ell \in [p]} \epsilon_{v^\ell} [Y_{[p]}] = \langle \{ \tau^0 [Y_0, I_0], \tau^{p-1} [I_{p-1}, Y_{p-1}] \} \cup \{ \tau^\ell [I_{\ell-1}, Y_\ell, I_\ell] : \ell \notin \{0, p-1\} \} \rangle_{[Y_{[p]}]}.$$

The TT multirank of 2 is furthermore minimal, since each matrifaction based on a partition of $[p]$ into non-empty sets has a matrix rank of 2. With respect to such partitions also the tensor $\epsilon_v [Y_{[p]}] + \left(\sum_{\ell \in [p]} \epsilon_{v^\ell} [Y_{[p]}] \right)$ has matrix ranks of 2. Since these are the only two activation tensors for the face $\mathcal{F}^{\triangleleft, 1}$.

Example 9 (Generalization of Example 8 to larger TT ranks). *We now generalize the construction of Example 8 by using the Hadamard distance $d(\cdot, \cdot)$ in $\{0, 1\}^p$, which counts the number of coordinates two vertices differ in. For $s \in \{1, \dots, p-1\}$ we define for a fixed $v[L] \in \{0, 1\}^p$ a polytope as the convex hull*

$$\{\tilde{v}[L] : d(\tilde{v}, v) \geq s\}.$$

The face to the normal $\frac{1}{2}(2v[L] - \mathbb{I}[L])$ is the convex hull

$$\mathcal{F}^{\triangleleft, s} := \text{conv}(\{\tilde{v}[L] : d(\tilde{v}, v) = s\})$$

containing $\binom{p}{s}$ vertices. We label these vertices by subsets $A \subset [p]$ of cardinality s and define for $\ell \in [p]$

$$v^A [L = \ell] = \begin{cases} v[\ell \in [p]] & \text{if } \ell \notin A \\ 1 - v[\ell \in [p]] & \text{if } \ell \in A \end{cases}$$

We now construct a TT with hidden variables I_ℓ and ranks $r_\ell = \min(\ell + 1, p - \ell + 1, s + 1)$ to represent the sum of their one-hot encodings. To this end, let there be TT cores for $\ell = 0$ by

$$\tau^0 [Y_0, I_0] = \epsilon_{v[L=0]} [Y_0] \otimes \epsilon_0 [I_0] + \epsilon_{1-v[L=0]} [Y_0] \otimes \epsilon_1 [I_0]$$

further for $\ell \notin \{0, p-1\}$ the cores

$$\tau^\ell [I_{\ell-1}, Y_\ell, I_\ell] = \sum_{\ell \in \{\max(s-p+\ell, 0), \dots, \max(\ell, s)\}} \epsilon_\ell [I_\ell] \otimes (\epsilon_{v[L=\ell]} [Y_\ell] \otimes \epsilon_\ell [I_{\ell+1}] + \epsilon_{1-v[L=\ell]} [Y_\ell] \otimes \epsilon_{\ell+1} [I_{\ell+1}])$$

and for $\ell = p-1$

$$\tau^{p-1} [I_{p-1}, Y_{p-1}] = \epsilon_0 [I_{p-1}] \otimes \epsilon_{1-v[L=p-1]} [Y_{p-1}] + \epsilon_1 [I_{p-1}] \otimes \epsilon_{v[L=p-1]} [Y_{p-1}].$$

Based on the interpretation, that the hidden variables I_ℓ count the Hamming distance of the vectors $v[L] \in \mathbb{R}^\ell \times 0_{p-\ell}$ and the respective $v^A [L] \in \mathbb{R}^\ell \times 0_{p-\ell}$ one can show that

$$\sum_{A \subset [p] : |A|=s} \epsilon_{v^A} [Y_{[p]}] = \langle \{ \tau^0 [Y_0, I_0], \tau^{p-1} [I_{p-1}, Y_{p-1}] \} \cup \{ \tau^\ell [I_{\ell-1}, Y_\ell, I_\ell] : \ell \notin \{0, p-1\} \} \rangle_{[Y_{[p]}]}.$$

Example 10 (Generalization of Example 9 to arbitrary HT formats). *Instead of aligning the Hamming count variables linearly, one can find a representation of the activation tensor*

$$\sum_{A \subset [p] : |A|=s} \epsilon_{v,A} [Y_{[p]}]$$

in an arbitrary directed acyclic tree hypergraph format $\mathcal{G}$, which hidden ranks are bounded by the number of leafs in the subtree and $s + 1$.

At any leaf of the tree we define a 2×2 matrix

$$\tau^\ell [Y_\ell, I_\ell] = \epsilon_{v[L=\ell]} [Y_\ell] \otimes \epsilon_0 [I_\ell] + \epsilon_{1-v[L=\ell]} [Y_\ell] \otimes \epsilon_1 [I_\ell] .$$

At each intermediate non-root hyperedge we choose an outgoing counting variable $I_{e^{\text{out}}}$ with dimension by $n_{e^{\text{out}}} = \min(\sum_{v \in e^{\text{in}}} n_{e^{\text{in}}}, s + 1)$ and define a tensor with the slices

$$\tau^e [I_{e^{\text{out}}}, I_{e^{\text{in}}}] = \beta^+ [Y_+ = i_{e^{\text{out}}}, I_{e^{\text{in}}}] .$$

We further build at the root hyperedge e a tensor

$$\tau^e [I_e] = \sum_{i : \langle i \rangle_{[\emptyset]} = s} \epsilon_{i_e} [I_e] .$$

Using the counting variable interpretation one can now show that

$$\sum_{A \subset [p] : |A|=s} \epsilon_{v,A} [Y_{[p]}] = \langle \{ \tau^e [I_e] : e \in \mathcal{E} \} \cup \{ \tau^\ell [Y_\ell, I_\ell] : v \in \mathcal{V} \} \rangle_{[Y_{[p]}]} .$$

5.4 Construction of Hybrid Logic Networks

We now constructively show, that any convex polytope with boolean vertices in $\mathbb{R}^p$ (a so called 0-1 polytope, see Ziegler (2000)) is the mean polytope of a family of Hybrid Logic Networks.

Theorem 7. *Let $\mathcal{M}$ an arbitrary polytope with boolean vertices in $\mathbb{R}^p$. Then we construct propositional formulas on atoms $X_{[p]}$ by*

$$f_0 [X_0] = \begin{cases} \top & \text{if } \mathcal{M}|_{\mathbb{R}^1 \times 0_{p-1}} = \{1\} \\ \perp & \text{if } \mathcal{M}|_{\mathbb{R}^1 \times 0_{p-1}} = \{0\} \\ X_0 & \text{if } \mathcal{M}|_{\mathbb{R}^1 \times 0_{p-1}} = [0, 1] \end{cases}$$

and iteratively for $\ell \in [p]$ with $\ell \geq 1$ by

$$f_\ell [X_{[\ell+1]}] = \bigwedge_{v[L] \in \mathcal{M}|_{\mathbb{R}^\ell \times 0_{p-\ell}} \cap \{0,1\}^p} \left(\left(\bigwedge_{\tilde{\ell} \in [\ell+1]} \neg^{1-v[L=\tilde{\ell}]} f_{\tilde{\ell}} [X_{[\tilde{\ell}]}] \right) \Rightarrow \begin{cases} \top & \text{if } \mathcal{M}|_{v \times \mathbb{R}^1 \times 0_{p-\ell-1}} = \{1\} \\ \perp & \text{if } \mathcal{M}|_{v \times \mathbb{R}^1 \times 0_{p-\ell-1}} = \{0\} \\ X_\ell & \text{if } \mathcal{M}|_{v \times \mathbb{R}^1 \times 0_{p-\ell-1}} = \{0, 1\} \end{cases} \right) .$$

Here we denote by $\mathcal{M}|_V$ the projections of the vertices in $\mathcal{M}$ onto the subspaces V , and by 0_p the zero vector in $\mathbb{R}^p$.

Proof. We show per induction, that for any $\ell \in [p]$ the family of Hybrid Logic Networks with the statistic $f_{[\ell+1]}$ by the first $\ell + 1$ formulas has the mean polytope

$$\mathcal{M}_{f_{[\ell+1]}, \mathbb{I}} = \mathcal{M}|_{\mathbb{R}^\ell \times 0_{p-\ell}} . \quad (2)$$

$\ell = 0$: The polytope $\mathcal{M}|_{\mathbb{R}^\ell \times 0_{p-\ell}} = \{1\}$ (respectively $\mathcal{M}|_{\mathbb{R}^\ell \times 0_{p-\ell}} = \{0\}$) is reproduced by f_0 being a tautology (respectively a contradiction). In the case $\mathcal{M}|_{\mathbb{R}^\ell \times 0_{p-\ell}} = [0, 1]$ the polytope is reproduced by the any formula, which is neither a tautology nor a contradiction, and the atomic formula X_0 is an example of such an contingency. Since the projection of a 0-1 polytope onto the first coordinates is itself a 0-1 polytope, these are the only possible cases and we conclude that in all

$$\mathcal{M}_{f_{[\ell+1]}, \mathbb{I}} = \mathcal{M}|_{\mathbb{R}^\ell \times 0_{p-\ell}} .$$

$\ell \rightarrow \ell + 1$: Let us assume, that (2) holds for a $\ell \in [p]$. Then for any $x_{[d]} \in \times_{k \in [d]} [m_k]$ there is exactly one $v[L] \in \mathcal{M}_{|\mathbb{R}^\ell \times 0_{p-\ell}|}$ such that $x_{[d]}$ is a model of

$$f_{\ell,v} := \bigwedge_{\tilde{\ell} \in [\ell+1]} \neg^{1-v[L=\ell]} f_{\ell} [X_{[\ell]}] .$$

This holds, since by the mutual contradiction of these formulas at most one can have $x_{[d]}$ as a model. Further, if none would have $x_{[d]}$ as a model, then the to $x_{[d]}$ corresponding vector of satisfactions $(f_{\ell} [X_{[d]} = x_{[d]}])_{\ell \in [\ell]}$ is not in $\mathcal{M}_{|\mathbb{R}^\ell \times 0_{p-\ell}|}$, which can not be the case. Now, the implications to all $\tilde{v}$ except for v are for the models $x_{[d]}$ of $f_{\ell,v}$ satisfied, and the satisfaction of f_{ℓ} thus only depends on the head of the implication to v . It follows, that the vertices of $\mathcal{M}_{f_{[\ell+2]}, \mathbb{I}}$ sharing the first ℓ coordinates with v are determined by the head of the implication at v . With the same arguments as in the case $\ell = 0$ we now notice, that in the three cases we construct vertex sets $v \times \{1\}$, $v \times \{1\}$ or $v \times \{0, 1\}$ if and only if they appear in the polytope $\mathcal{M}_{|\mathbb{R}^{\ell+1} \times 0_{p-\ell-1}|}$. This establishes for each vertex v of $\mathcal{M}_{f_{[\ell+1]}, \mathbb{I}}$ that

$$(\mathcal{M}_{f_{[\ell+2]}, \mathbb{I}})|_{v \times \mathbb{R} \times 0_{p-\ell-1}} = (\mathcal{M}_{|\mathbb{R}^\ell \times 0_{p-\ell}|})|_{v \times \mathbb{R} \times 0_{p-\ell-1}} .$$

Since for any vertex in $\mathcal{M}$ we find a unique vertex in $\mathcal{M}_{f_{[\ell+1]}, \mathbb{I}}$ sharing the first ℓ coordinates, we have that (2) holds for $\ell + 1$.

By induction the equation (2) holds for arbitrary $\ell \in [p]$. For $\ell = p - 1$ the equation is the claim. $\square$

5.5 Further Examples of non-cubelike polytopes

Example 11 (Ising model on 2 nodes). Consider two boolean variables X_0, X_1 and the Ising statistic by three propositional formulas

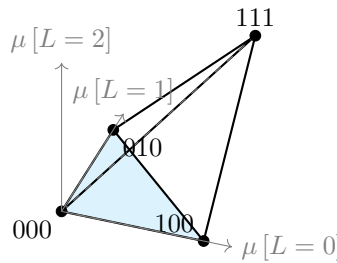
$$f_0 [X_{[2]}] = X_0 \quad , \quad f_1 [X_{[2]}] = X_1 \quad \text{and} \quad f_2 [X_{[2]}] = X_0 \wedge X_1 .$$

In the Ising interpretation, the boolean variables represent interacting spins at two locations. Their value is then measured by the first two formulas and their interaction by the third.

The vertices of the mean polytope to this statistic are

$$t(x_0, x_1) = [f_0 [X_{[2]} = x_{[2]}], f_1 [X_{[2]} = x_{[2]}], f_2 [X_{[2]} = x_{[2]}]]^T = \begin{cases} [0, 0, 0]^T & \text{if } (x_0, x_1) = (0, 0) \\ [0, 1, 0]^T & \text{if } (x_0, x_1) = (0, 1) \\ [1, 0, 0]^T & \text{if } (x_0, x_1) = (1, 0) \\ [1, 1, 1]^T & \text{if } (x_0, x_1) = (1, 1) \end{cases}$$

and the mean polytope is the convex hull of these, sketched as:



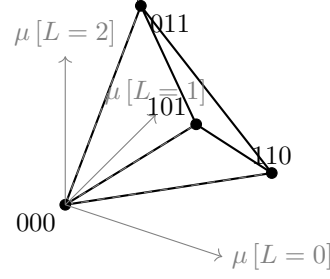
We notice, that the convex hull of any three out of the four vertices builds a facet. The only cube-like facet of these four is the convex hull of $\{[0, 0, 0]^T, [1, 0, 0]^T, [0, 1, 0]^T\}$ (sketched in cyan in the above plot) to which we have the parametrization by $(\mathcal{U}, y_{\mathcal{U}}) = (\{2\}, 0)$.

As a consequence, there are maximum entropy distributions of mean parameters in the Ising statistics, which do not have a representation by an elementary CompActNet in the Ising statistic.

Example 12 (Codeword Polytopes). Consider the codeword polytope with the vertices

$$\{y_{[3]} : \sum_{k \in [3]} y_k = 0\}$$

We interpret the vertices by the strings of length 3, which parity vanishes. Such constrained are common in the construction of codewords using parity-check codes (see Gallager (1963)).



All facets are non-cube-like, whereas all other faces are cube-like (the edges are labeled by the 6 facets of $[0, 1]^3$).

Each facet is characterized by the Hamming distance of exactly 1 to a single of the 4 cube vertices, which are not codewords (see Example 3.9 in Wainwright and Jordan (2008)). We can construct TT activation tensors with rank 2, which are calculating in the hidden ranks the Hamming distance to these vertex. Each core tensor in that format adds the contribution of a mode to the Hamming distance.

In more generality, Low-Density-Parity-Checking Codes (see Gallager (1963)) construct codewords by a collection of parity constraints on subsets of variables. Such polytopes are also known as cycle polytopes, and studied e.g. in Grötschel et al. (1993).

6 Outlook

This is relevant for Parameter Estimation: Instabilities when fitting a mean parameter on a non cube-like face.

Further theory to be investigated:

- Face representability of randomly drawn polytopes (e.g. expectation of a face rank).
- Asymptotic properties for large polytopes (does the average face rank concentrate at an expectation)?

Investigate the connection with convex polytope investigations:

- Facet complexity (the amount of information to specify the face by inequalities) Grötschel et al. (1993)

Border-rank development to resolve the issue of non-elementary faces: When defining a boarder rank of the activation tensors by limiting CompActNets, then any maximum entropy distribution has border rank 1.

Application in NeSy: By format selection (i.e. which complexity is required to represent a face).

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A TT Face Activation

A.1 Generic Construction

When the vertices are characterized by the sum of local terms, we can construct the face activation tensor in TT format, by summation cores. This is for example the case for the examples constructed based on Hamming distances.

A.2 Relation with Facet Complexity

Facet complexity (Grötschel et al., 1993, Definition 6.2.2) bounds the encoding length of the inequalities describing a polytope.

For any face $\mathcal{F}$ we find a normal $a \in \mathbb{R}^p$ and $b \in \mathbb{R}$ such that x

$$\mathcal{F} = \{\mu[L] \in \mathcal{M} : \langle a[L], \mu[L] \rangle_{[\emptyset]} = b\}.$$

We can thus represent a face activating tensor by computing the contraction $\langle a[L], y[L] \rangle_{[\emptyset]}$.

To this end, let $\mathcal{U} \subset \mathbb{R}$ be a set with index interpretation function

$$I : [r] \rightarrow \mathcal{U}.$$

We demand $0, b \in \mathcal{U}$. We can for example choose $\mathcal{U}$ as the d -bit binary integer representation.

We then define for $\ell \in [p]$ the summation operations

$$+^{\ell, a[L=\ell]} : \mathcal{U} \times [n] \rightarrow \mathcal{U}$$

which satisfies for any $u \in \mathcal{U}$ and y_ℓ for which $u + a[L=\ell] \cdot y_\ell \in \mathcal{U}$ the equation

$$+^{\ell, a[L=\ell]}(u, y_\ell) = u + a[L=\ell] \cdot y_\ell.$$

We further assume, that for any tuple $y_{[p]} \in \times_{\ell \in [p]} [n_\ell]$ and any $\tilde{\ell} \in [p]$

$$\sum_{\ell \leq \tilde{\ell}} a[L=\ell] \cdot y_{\tilde{\ell}} \in \mathcal{U}.$$

Then we have

$$\sum_{\mu \in \mathcal{N}_{i,v}^{\mathcal{F}}} \epsilon_\mu [Y_{[p]}] = \left\langle \{\beta^{+, \ell, a[L=\ell]} [Y_{+^{\ell, a[L=\ell]}}, Y_\ell] : \ell \in [p]\} \cup \{\epsilon_{I^{-1}(0)} [Y_{+^{-1, a[L=-1]}}, \epsilon_{I^{-1}(b)} [Y_{+^{p-1, a[L=p-1]}}, \dots]\} \right\rangle_{[Y_{[p]}]}.$$

The variable $Y_{+^{-1, a[L=-1]}}$ is auxiliary to initialize the sum with 0.

We can weaken the assumptions on $\mathcal{U}$: When $a[L]$ is positive, then any partial sum which exceeds b can be represented by any member of $\mathcal{U}$ which is larger than b . This brings further sparsity, since no values larger than b are needed to represent the face (add for example the ∞ element to $\mathcal{U}$ and represent any partial sum exceeding b by it).

Definition 10. We say that a face $\mathcal{F}$ has TT face complexity r if there exists a normal $a \in \mathbb{R}^p$ and $b \in \mathbb{R}$ with

$$\mathcal{F} = \{\mu[L] \in \mathcal{M} : \langle a[L], \mu[L] \rangle_{[\emptyset]} = b\}.$$

and there is a subset $\mathcal{U}$ of cardinality at most r , for which $0, b \in \mathcal{U}$ and for any tuple $y_{[p]} \in \times_{\ell \in [p]} [n_\ell]$ and any $\tilde{\ell} \in [p]$

$$\sum_{\ell \leq \tilde{\ell}} a[L = \ell] \cdot y_{\tilde{\ell}} \in \mathcal{U}.$$

Comparison with facet complexity (Grötschel et al., 1993, Definition 6.2.2):

- We define complexity here without the log (which would get the string length), to make a direct connection with TT ranks.
- Facet complexity considers only the representation of the equation, not the demand of computing it.
- Facet complexity considers just facets, not general faces, since it makes a statement about encoding the halfspace representation of the whole polytope.

Theorem 8. Let $\mathcal{F}$ be a face with TT face complexity r . Then the face activation tensor has a TT rank of at most r .

Proof. See above construction. □